

Chapter 0

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Throughout, V is a finite-dimensional vector space over a field F , and $T : V \rightarrow V$ is linear (equivalently $A \in M_n(F)$ after a choice of basis). Write $p_T(x) = \det(xI - T)$ for the characteristic polynomial and $m_T(x)$ for the minimal polynomial of T .

Chapter 1

Diagonalization

Definition 1.1. T is *diagonalizable* if V has a basis consisting of eigenvectors of T . For $\lambda \in F$, the *eigenspace* is $E_\lambda = \ker(T - \lambda I)$.

Lemma 1.2. Eigenvectors corresponding to distinct eigenvalues of T are linearly independent.

Proof: Let $v_1, \dots, v_n \in V$ be the smallest set of linearly dependent eigenvectors of T such that $Tv_i = \lambda_i v_i$ such that $\lambda_i \neq \lambda_j \Leftrightarrow i \neq j$.

$$\begin{aligned} \sum_{i=1}^n a_i v_i &= 0 \\ \Rightarrow T \sum_{i=1}^n a_i v_i &= \sum_{i=1}^n a_i \lambda_i v_i = 0 \quad (\text{E1}) \end{aligned}$$

$$\lambda_1 \sum_{i=1}^n a_i v_i = 0 \quad (\text{E2})$$

$$0 = \text{E1} - \text{E2} = \sum_{i=1}^n (\lambda_i - \lambda_1) a_i v_i = 0$$

We now get a smaller set $\{v_2, \dots, v_n\}$ of linearly dependent eigenvectors and hence we reach a contradiction.

□

Lemma 1.3. For distinct eigenvalues $\lambda_1, \dots, \lambda_k$ of T , the sum $E_{\lambda_1} + \dots + E_{\lambda_k}$ is direct.

Lemma 1.4. For every eigenvalue λ of T , $\dim E_\lambda \leq \text{mult}_{p_T}(\lambda)$, the multiplicity of λ as a root of p_T . Thus the algebraic multiplicity of a eigenvalue is greater than or equal to the geometric multiplicity of the eigenvalue

Proof: Let $\dim E_\lambda = g$ and $\dim V = n$. Let $v_1, \dots, v_g$ be a basis for $\dim E_\lambda$. Extend the basis to V , thus let $v_1, \dots, v_g, w_1, \dots, w_{n-g}$ be a basis for V . Let $P : V \rightarrow \mathbb{F}^n$ be the linear isomorphism where $v_i \rightarrow e_i$ and $w_j \rightarrow e_{g+j}$. Thus note that $PTP^{-1}e_i = \lambda e_i \Leftrightarrow i = 1, \dots, g$. Thus

$$PTP^{-1} = \begin{pmatrix} \lambda I & B \\ 0 & C \end{pmatrix}$$

where I is the $g \times g$ identity matrix. $B \in \text{Mat}_{g \times (n-g)}(\mathbb{F})$ and $D \in \text{Mat}_{(n-g) \times (n-g)}(\mathbb{F})$. Note that $p_T(x) = p_{PTP^{-1}}(x)$.

$$\begin{aligned} p_{PTP^{-1}}(x) &= \det \begin{pmatrix} (\lambda - x)I & B \\ 0 & C - xI \end{pmatrix} \\ &= \det((\lambda - x)I) \det(C - xI) \\ &= (\lambda - x)^g \det(C - xI) = p_T(x) \end{aligned}$$

Thus $(\lambda - x)^g \mid p_T(x)$, thus $(\lambda - x)$ is a factor of $p_T(x)$ furthermore g is less than or equal to the multiplicity of λ as a root of p_T . $\square$

Lemma 1.5 (That Kid's Lemma). Let U, V, W be $\mathbb{F}$ -vector spaces. Let $A \in \text{Hom}_{\mathbb{F}}(U, V)$ and $B \in \text{Hom}_{\mathbb{F}}(V, W)$. Then

$$\dim \ker BA \leq \dim \ker A + \dim \ker B$$

Proof: Note that $\ker A \subseteq \ker BA$ and note that $A(\ker BA) \subseteq \ker B$. By rank nullity we have the following:

$$\begin{aligned} \dim \ker BA &= \dim \ker(A|_{\ker BA}) + \dim \text{im}(A|_{\ker BA}) \\ &= \dim \ker A + \dim \text{im}(A|_{\ker BA}) \\ &\leq \dim \ker A + \dim \ker B \end{aligned}$$

$\square$

Theorem 1.6 (Diagonalizability Criterion). The following are equivalent:

- (a) T is diagonalizable.
- (b) $V = \bigoplus_{\lambda} E_{\lambda}$, the sum over all eigenvalues of T .
- (c) $\sum_{\lambda} \dim E_{\lambda} = \dim V$.
- (d) $m_T(x)$ factors as a product of distinct linear factors over F .

Proof: $a \Leftrightarrow b \Leftrightarrow c$: Tautological using the definition and Lemma 1.2.

$a \Rightarrow d$: Suppose $T : V \rightarrow V$ is diagonalizable. Thus $v_1, \dots, v_n \in V$ a basis V consisting of eigenvectors for T . Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues with multiplicities $k_1, \dots, k_p$. Thus

$$Tv_r = \lambda_p v_r \text{ where } K_{p-1} = \sum_{i=1}^{p-1} k_i \leq r < \sum_{i=1}^p k_i = K_p.$$

Note that

$$(T - \lambda_p I)v_r = 0$$

for $K_{p-1} \leq r < K_p$. Thus

$$\prod_{i=1}^m (T - \lambda_i I)v_i = 0$$

for all $i = 1, \dots, n$. Thus

$$m_T(x) \mid \prod_{i=1}^m (x - \lambda_i) = s(x).$$

Since $s(x)$ consists of distinct linear factors so does $m_T(x)$.

$d \Rightarrow b \Rightarrow a$: Suppose $m_T(x)$ consists of distinct linear factors. We will first do a quick lemma: for $A \in \text{Hom}_{\mathbb{F}}(V, W)$ and $B \in \text{Hom}_{\mathbb{F}}(U, V)$

$$\dim \ker AB \leq \dim \ker A + \dim \ker B.$$

Note that $\forall k \in \ker AB$, $B(k) \in \ker A$. Thus $B|_{\ker AB} : \ker AB \rightarrow \ker A$. Apply rank nullity theorem:

$$\dim \ker AB = \dim \ker(B|_{\ker AB}) + \dim \text{im}(B|_{\ker AB}) \leq \dim \ker(B|_{\ker AB}) + \dim \ker A.$$

Note that $\ker AB \cap \ker B = \ker B$. Thus $\ker(B|_{\ker AB}) = \ker B$. Thus

$$\dim \ker AB \leq \dim \ker B + \dim \ker A.$$

Hence we have the lemma. Now

$$m_T(x) = \prod_{i=1}^m (x - \lambda_i)$$

where $\lambda_1, \dots, \lambda_m$ are distinct. Note that $m_T(T) = 0$. Thus

$$\dim V = \dim \ker m_T(T) \leq \sum_{i=1}^n \dim \ker (T - \lambda_i I) = \sum_{i=1}^m \dim E_{\lambda_i}.$$

Since $E_{\lambda_i} \cap E_{\lambda_j} = \{0\}$ for $i \neq j$,

$$\sum_{i=1}^m \dim E_{\lambda_i} = \dim \bigoplus_{i=1}^m E_{\lambda_i} \leq \dim V.$$

Thus we have equality and we get c which implies diagonalizable. $\square$

Lemma 1.7 (Idempotent Decomposition). If $m_T(x) = \prod_{i=1}^k (x - \lambda_i)$ with the λ_i distinct, there exist polynomials $e_1, \dots, e_k \in F[x]$ such that $e_i(T)$ is the projection of V onto E_{λ_i} along $\bigoplus_{j \neq i} E_{\lambda_j}$, with $\sum_i e_i(T) = I$ and $e_i(T)e_j(T) = 0$ for $i \neq j$.

Proof: Let

$$e_i(x) = \prod_{\substack{1 \leq j \leq k \\ i \neq j}} \frac{x - \lambda_j}{\lambda_i - \lambda_j}.$$

Note the following

$$\begin{aligned} e_i(\lambda_i) &= \prod_{\substack{1 \leq j \leq k \\ i \neq j}} \frac{\lambda_i - \lambda_j}{\lambda_i - \lambda_j} = 1 \\ e_i(\lambda_j) &= 0 \end{aligned}$$

Thus for $v \in E_{\lambda_j}$ for $i \neq j$ and $w \in E_{\lambda_i}$. Consider

$$\begin{aligned} (T - \lambda_m I)v &= Tv - \lambda_m v = (\lambda_j - \lambda_m)v \\ (T - \lambda_m I)w &= (\lambda_i - \lambda_m)w \\ e_i(T)v &= \left(\prod_{\substack{1 \leq j \leq k \\ i \neq j}} \frac{T - \lambda_j I}{\lambda_i - \lambda_j} \right) v \\ &= \left(\prod_{\substack{1 \leq m \leq k \\ m \neq i}} \frac{\lambda_j - \lambda_m}{\lambda_i - \lambda_j} \right) v = e_i(\lambda_j)v = 0 \\ e_i(T)w &= \left(\prod_{\substack{1 \leq j \leq k \\ i \neq j}} \frac{T - \lambda_j I}{\lambda_i - \lambda_j} \right) w \\ &= e_i(\lambda_i)w = w \end{aligned}$$

Thus

$$e_i(T)|_{E_a} = \begin{cases} I & \text{if } a = i \\ 0 & \text{otherwise} \end{cases}$$

Thus

$$\sum_{i=1}^k e_i(T) = I \text{ for } \bigoplus_{i=1}^k E_{\lambda_i} = V$$

□

Lemma 1.8 (Invariance of Diagonalizability Under Restriction). If T is diagonalizable and $W \leq V$ is a T -invariant subspace, then $T|_W$ is diagonalizable.

Proof: It suffices to show that $m_{T|_W} \mid m_T$, thus since m_T has distinct linear factors so does $m_{T|_W}$, and hence $T|_W$ is diagonalizable. To do this we will show that

$$m_T(T|_W) = 0.$$

Note the following by definition $(T|_W)w = Tw$ for all $w \in W$. Thus $(T|_W)^n w = T^n w \Rightarrow m_T(T|_W)w = m_T(T)w = 0$. Thus $m_T(T|_W) : W \rightarrow W$ is the 0 map and hence $m_{T|_W} \mid m_T$. $\square$

Theorem 1.9 (Simultaneous Diagonalization). If $S, T : V \rightarrow V$ are each diagonalizable and $ST = TS$, then there is a basis of V consisting of vectors that are simultaneously eigenvectors of S and of T .

Proof: Let E_{λ_i} for $i = 1, \dots, k$ be the eigenspaces of S . $\forall v \in E_{\lambda_i}$ note the following:

$$STv = T Sv = \lambda_i T v \Rightarrow T v \in E_{\lambda_i}$$

Thus E_{λ_i} for all i are T invariant subspaces. Hence $T|_{E_{\lambda_i}}$ is diagonalizable and hence you can find a basis of eigenvectors in E_{λ_i} (for all i) that is a eigenbasis for both $S|_{E_{\lambda_i}}$ and $T|_{\lambda_i}$. Combine the basis for all E_{λ_i} to get a basis for V . The basis is a eigenbasis for S and T . $\square$

Chapter 2

The Cayley–Hamilton Theorem

Theorem 2.1 (Cayley–Hamilton). $p_T(T) = 0$.

Proof: We will first prove this theorem is true over $\mathbb{C}$, then we will prove over any arbitrary field.

1. **Diagonal Matrices:** Let $D = \text{diag}(\lambda_1, \dots, \lambda_n)$. Then

$$\begin{aligned} p_D(x) &= \det(D - xI) \\ &= \det(\text{diag}(\lambda_1 - x, \dots, \lambda_n - x)) \\ &= \prod_{i=1}^n (\lambda_i - x). \end{aligned}$$

Thus

$$\begin{aligned} p_D(D) &= \prod_{i=1}^n (\lambda_i I - D) \\ &= \prod_{i=1}^n \text{diag}(\lambda_1 - \lambda_i, \dots, \lambda_n - \lambda_i) \\ &= 0 \end{aligned}$$

2. **Diagonalizable Linear Transformations:** Let $T : V \rightarrow V$ be a diagonalizable linear Transformation, thus $\exists P : V \rightarrow \mathbb{F}^n$ a linear isomorphism and $D \in \text{Mat}_{n \times n}(\mathbb{F})$ a diagonal matrix where $T = P^{-1}DP$. Note $T^k = PD^kP^{-1}$ for all $k = 0, 1, 2, \dots$. Thus since p_T is polynomial.

$$\begin{aligned} p_T(x) &= \det(T - xI) \\ &= \det(P^{-1}DP - \lambda I) \\ &= \det(P^{-1}(D - \lambda I)P) \\ &= \det(P)^{-1} p_D(x) \det(P) = p_D(x) \end{aligned}$$

$$p_T(T) = p_D(T) = p_D(P^{-1}DP) = P^{-1}p_D(D)P = 0$$

3. **All Matrices:** Diagonalizable linear Transformations are dense over all $\text{End}(V)$ for V a vector space over $\mathbb{C}$. Since $A \rightarrow p_A(A)$ is a continuous map from $\text{End}(V) \rightarrow \mathbb{C}$ which is 0 on a dense set and hence 0 identically.

Thus we have proven for all $\text{End}(V)$ for V a $\mathbb{C}$ vector space the statement is true.

Let $R = \mathbb{Z}[x_{11}, \dots, x_{nn}]$. Let $M = [x_{ij}]_{1 \leq i, j \leq n} \in \text{Mat}_{n \times n}(R)$. Note that the coefficients of p_M are in R (ie polynomials in the variables $x_{11}, \dots, x_{nn}$). Thus when we evaluate $p_M(M)$, each entry will be a polynomial in $\mathbb{Z}[x_{11}, \dots, x_{nn}]$. Let P_{ij} denote the polynomial in the (i, j) -entry of $p_M(M)$. We want to show that $P_{ij} = 0$ for all $1 \leq i, j \leq n$. Note that by 1 - 3, $\forall A \in \text{Mat}(\mathbb{C})$, we have that $p_A(A) = 0$. Thus $\forall \vec{a} \in \mathbb{C}^{n^2}$, the matrix $A = M(\vec{a}) \Rightarrow p_A(A) = p_{M(\vec{a})}(M(\vec{a})) = 0$ thus by definition $P_{ij}(\vec{a}) = 0$. Thus since $P_{i,j}(\vec{a}) = 0$ for all $\vec{a} \in \mathbb{C}^{n^2}$, which means that $P_{ij} \equiv 0$. Thus $p_M(M) = 0$. Let $\mathbb{F}$ be an arbitrary field. Let $B = [B_{ij}] \in \text{Mat}_{n \times n}(\mathbb{F})$, define the evaluation map:

$$\begin{aligned} \Phi : \mathbb{Z}[x_{11}, \dots, x_{nn}] &\rightarrow \mathbb{F} \\ x_{ij} &\rightarrow B_{ij}. \end{aligned}$$

$$0 = \Phi(P_{ij}) = (p_B(B))_{ij} \Rightarrow p_B(B) = 0$$

□

Lemma 2.2 (Adjugate Identity). Let $B(x) = \text{adj}(xI - T) \in M_n(F[x])$. Then

$$(xI - T)B(x) = p_T(x)I.$$

Lemma 2.3 (Triangularizability). If $p_T(x)$ splits into linear factors over F , then there is a basis of V in which T is represented by an upper triangular matrix.

Corollary 2.4. $m_T(x) \mid p_T(x)$.

Corollary 2.5. $m_T(x)$ and $p_T(x)$ have the same roots in $\overline{F}$ (equivalently, the same irreducible factors over F), possibly with different multiplicities.

Chapter 3

Jordan Canonical Form

Assume in this section that $p_T(x)$ splits over $\mathbb{F}$.

Definition 3.1. For $\lambda \in F$, the *generalized eigenspace* is $K_\lambda = \ker(T - \lambda I)^{\dim V}$. An operator N is *nilpotent* if $N^r = 0$ for some $r \geq 1$. The *Jordan block* $J_k(\lambda) \in M_k(F)$ is the upper triangular matrix with λ on the diagonal and 1 on the superdiagonal.

Theorem 3.2 (Primary Decomposition Theorem). If $m_T(x) = \prod_{i=1}^r q_i(x)^{e_i}$ with $q_i \in F[x]$ distinct monic irreducibles, then

$$V = \bigoplus_{i=1}^r \ker q_i(T)^{e_i},$$

and each summand is T -invariant.

Proof:

1. **Isomorphism :** Let $v_1, \dots, v_k$ be the smallest set of nonzero linearly dependent vectors such that $v_i \in \ker q_i(T)^{e_i}$. Thus there exists $a_1, \dots, a_k \in \mathbb{F}$ such that $(a_1, \dots, a_k) \neq 0$ and

$$\begin{aligned} \sum_{i=1}^k a_i v_i &= 0 \\ q_1(T)^{e_1} \left(\sum_{i=1}^k a_i v_i \right) &= 0 \\ \sum_{i=2}^k a_i q_1(T)^{e_i} v_i &= 0 \end{aligned}$$

Let $w_i = q_1(T)^{e_1} v_i$. $q_i(T)^{e_i} w_i = q_i(T)^{e_i} (q_1(T)^{e_1} v_i) = 0$ since polynomials in T commute. Hence $w_i \in \ker q_i(T)^{e_i}$. Distinct irreducible polynomials are coprime, hence by Bezout's Theorem there $\exists a_i, b_i \in \mathbb{F}[x]$ where

$$1 = a_i(x) q_i(x)^{e_i} + b_i(x) q_1(x)^{e_1}$$

thus

$$I = a_i(T) q_i(T)^{e_i} + b_i(T) q_1(T)^{e_1} \Rightarrow v_i = b_i(T) q_1(T)^{e_1} v_i \Rightarrow v_i = b_i(T) w_i \Rightarrow w_i \neq 0.$$

Thus

$$\sum_{i=2}^n a_i w_i = 0$$

is a shorter sum of nonzero linearly dependent vectors. Furthermore $a_i \neq 0$ for all $i \in \{2, \dots, n\}$, because that would imply $a_1 v_1 = 0 \Rightarrow a_1 = 0$ or $v_1 = 0$ both of which introduce contradictions.

By definition, $m_T(T) = 0$. Hence $\dim \ker m_T(T) = \dim V$. By Lemma 3.5, note that

$$\dim V = \dim \ker m_T(T) \leq \sum_{i=1}^n \dim \ker q_i(T)^{e_i} \leq \dim V.$$

Thus

$$V = \bigoplus_{i=1}^r \ker q_i(T)^{e_i}$$

2. **T -invariance:** Polynomials in T commute. Hence $q_i(T)^{e_i} T = T q_i(T)^{e_i}$. This directly implies T invariance (apply then commute).

□

Theorem 3.3 (Generalized Eigenspace Decomposition). Write $m_T(x) = \prod_{\lambda} (x - \lambda)^{e_{\lambda}}$. Then

$$V = \bigoplus_{\lambda} K_{\lambda},$$

and $\dim K_{\lambda} = \text{mult}_{p_T}(\lambda)$ for each eigenvalue λ .

Proof: Directly apply Theorem 3.2 to get the isomorphism. Since each K_{λ} is T invariant, define $T_{\lambda} = T|_{K_{\lambda}}$. Since the total space is a direct product of the T invariant subspaces, you can define the characteristic polynomial as follows:

$$p_T(x) = \prod_{\lambda} p_{T_{\lambda}}(x).$$

We will study $p_{T_{\lambda}}(x)$. Note that $K_{\lambda} := \ker (T - \lambda I)^{\dim V}$. Thus by definition on K_{λ} , $(T_{\lambda} - \lambda I)$ is a Nilpotent operator, hence its only eigenvalue is 0. Thus T_{λ} only has a eigenvalue of λ . The characteristic polynomial must have degree equal to the dimension of the domain, so

$$p_{T_{\lambda}}(x) = (x - \lambda)^{\dim K_{\lambda}}.$$

Thus

$$p_T(x) = \prod_{\lambda} (x - \lambda)^{\dim K_{\lambda}} \Rightarrow \dim K_{\lambda} = \text{mult}_{p_T}(\lambda)$$

□

Theorem 3.4 (Nilpotent Structure Theorem). If $N : W \rightarrow W$ is nilpotent, there is a basis of W in which N is block diagonal with blocks $J_{k_1}(0), \dots, J_{k_s}(0)$. The multiset $\{k_1, \dots, k_s\}$ is uniquely determined by N .

Proof: Suppose N is nilpotent (and wlog $N \neq 0$ otherwise theorem trivializes). That means $N^r = 0$ for some $r \geq 1$. Note the following $\ker N^k \subseteq \ker N^{k+1}$ for all $k \geq 0$. Furthermore, you know that $\forall k \geq 0$ we have that $\ker N^k \subseteq W$. So you have a nondecreasing sequence of the dimensions of the kernels of N^k which has a upper bound of the dimension of W . Furthermore, by the fact that N is nilpotent you know that $N^r = 0$ for some r , hence $W = \ker N^r$. Hence there is a minimal r such that $\ker N^r = W$ but $\ker N^{r-1} \neq W$.

The minimal polynomial $m_N(x) \mid x^r$, thus the only eigenvalue of N is 0. To prove this statement we will induct on the dimension of our vector space. For the base case, let for any 1 dimensional vector space. $N : W \rightarrow W$ must be just be scalar multiplication ie $v \rightarrow \lambda v$. As proven above, the only eigenvalue of N is 0, so $\lambda = 0$. Hence $N = J_1(0)$. Assume the statement is true for all nilpotent linear Transformations acting on vector spaces of dimension less than or equal to n . Let N be a nilpotent linear transformation acting on W where $\dim W = n + 1$. Fix a $w_1 \in \ker N^r / \ker N^{r-1}$. Assume the contrary that $w_1 = Nu \Rightarrow N^{r-1}w_1 = N^ru = 0$, which is a contradiction (hence $w_1 \notin \text{im } N$ which will come up later). Consider the set of vectors $\{N^j w_1 : 0 \leq j \leq r\}$. We want to show that this is linearly independent. Let $0 \leq j_1 < \dots < j_k \leq r$ be the smallest set of indices such that

$$\sum_{i=1}^k a_i N^{j_i} w_1 = 0$$

and $a_i \neq 0$. Let $m = s_1 - j_k + 1$. Then

$$\begin{aligned} N^m \sum_{i=1}^k a_i N^{j_i} w_1 &= 0 \\ \sum_{i=1}^{k-1} a_i N^{j_i+m} w_1 &= 0 \end{aligned}$$

Note that $N^{j_i+m} w_1 \neq 0$ for $i \in \{1, \dots, k-1\}$, by definition. Thus we got a smaller set of indices which are linearly dependent and a contradiction occurs. Thus $\{N^j w_1 : 0 \leq j < r\}$ are linearly independent vectors. Note that the subspace generated by these vectors are N -invariant, and $N(N^j w_1) = N^{j+1} w_1$, so on this subspace N looks like $J_r(0)$. Let $W_1 = \text{span}(w_1, Nw_1, \dots, N^{r-1}w_1)$

Consider the quotient space W/W_1 . Note that $N(W_1) \subset W_1$. Since W_1 is N -invariant, N induces a nilpotent linear map, which we will call $\tilde{N}$, on the quotient space where $v + W_1 \rightarrow N(v) + W_1$. Note that $\dim W_1 \geq 1 \Rightarrow \dim(W/W_1) \leq \dim W - 1 = n - 1$, hence by the inductive assumption there exists a basis of disjoint cyclic subspaces. We want to “pull back”

the basis from the quotient space to get a appropriate basis for W , while maintaining the appropriate block structure. By assumption we have

$$\tilde{N}^{m+1}(u + W_1) = N^{m+1}u + W_1 = W_1.$$

Thus

$$\begin{aligned} N^{m+1}u &= \sum_{i=0}^{r-1} a_i N^i w_1 \\ N^r u &= N^{r-m-1} \sum_{i=0}^{r-1} a_i N^i w_i \\ &= \sum_{i=0}^{r-1} a_i N^{r+i-m-1} w_i \end{aligned}$$

Note for $i = \{m+1, \dots, r-1\}$, $N^{r+i-m-1} = N^r = 0$, but for $i \leq m$, $r+i-m-1 \leq r-1$. Hence

$$0 = N^r u = \sum_{i=1}^m a_i N^{r+i-m-1} w_1.$$

This implies that $a_i = 0$ for $1 \leq i \leq m$. Hence

$$N^{m+1}u = \sum_{i=m+1}^{r-1} a_i N^i w_1 \Rightarrow N^{m+1} \left(u - \sum_{i=1}^{r-m-2} a_{i+m+1} N^i w_1 \right) = 0.$$

Let

$$\begin{aligned} x &= \sum_{i=1}^{r-m-2} a_{i+m+1} N^i w_i \\ u' &= u - x. \end{aligned}$$

Thus the sequence $u', Nu', \dots, N^{m+1}u' = 0$ exactly when it supposed to. Suppose $N^k u' \in W_1$ for $1 \leq k \leq m$. Then $\tilde{N}^k(u' + W_1) = \tilde{N}^k(u + W_1) = 0$ which contradicts the maximality of m . Thus $W_1 \cap W_2 = \{0\}$. Thus we can keep peeling off dimensions off of our vector space W and using the inductive hypothesis on

$$W / \bigoplus_i^k W_i$$

to get W_{k+1} subspace we we find to be disjoint from all ther subspaces and by the inductive hypothesis to have a cyclic structure. Now we need to prove uniqueness of the multiset of block sizes. Note the following fact: $\dim \ker N^k - \dim \ker N^{k-1}$ is the number of blocks of at least size k for $k \in \mathbb{N}$. Then $-\dim \ker N^{k+1} + 2 \dim \ker N^k - \dim \ker N^{k-1}$ is the number of blocks of size k for $k \in \mathbb{N}$. Note these numbers have nothing to do with the choice of basis but are invariants of the transformation itself.

□

Lemma 3.5. If $N : W \rightarrow W$ is nilpotent and $W \neq 0$, then $\text{im } N$ is a proper, N -invariant subspace of W , and $N|_{\text{im } N}$ is nilpotent.

Proof: Suppose $W \neq 0$ and $N : W \rightarrow W$ is nilpotent. $\forall v \in \text{im } N$, $Nv \in \text{im } N$. Thus $\text{im } N$ is a N -invariant subspace by definition. Furthermore, since $N^r = 0$ for some $r \geq 1$. We know that $N^r v = (N|_{\text{im } N})^r v = 0$. Thus $(N|_{\text{im } N})^r = 0$ and $N|_{\text{im } N}$ is nilpotent. □

Theorem 3.6 (Jordan Canonical Form: Existence). There is a basis of V in which the matrix of T is block diagonal, each block a Jordan block $J_k(\lambda)$ for some eigenvalue λ of T . This block-diagonal matrix is unique up to permutation of the blocks.

Theorem 3.7 (Jordan Canonical Form: Uniqueness Formula). For an eigenvalue λ and $k \geq 1$, the number of Jordan blocks $J_k(\lambda)$ in the Jordan form of T equals

$$\text{rank}(T - \lambda I)^{k-1} - 2\text{rank}(T - \lambda I)^k + \text{rank}(T - \lambda I)^{k+1}.$$

Proof: Apply Theorem 3.3 and Theorem 3.4, straight forward. □

Chapter 4

Rational Canonical Form

F is an arbitrary field. Regard V as an $F[x]$ -module via $x \cdot v := Tv$.

Definition 4.1. For $v \in V$, the T -cyclic subspace generated by v is $Z(v, T) = \text{span}\{v, Tv, T^2v, \dots\}$. The T -annihilator of v is the monic generator $\text{ann}(v)$ of the ideal $\{f \in F[x] : f(T)v = 0\}$. For a monic polynomial $f(x) = x^d + c_{d-1}x^{d-1} + \dots + c_0$, the companion matrix $C(f)$ is the $d \times d$ matrix

$$C(f) = \begin{pmatrix} 0 & 0 & \dots & 0 & -c_0 \\ 1 & 0 & \dots & 0 & -c_1 \\ 0 & 1 & \dots & 0 & -c_2 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \dots & 1 & -c_{d-1} \end{pmatrix}.$$

Lemma 4.2 (Cyclic Basis Lemma). If $V = Z(v, T)$ and $\text{ann}(v)$ has degree d , then $\{v, Tv, \dots, T^{d-1}v\}$ is a basis of V , and the matrix of T in this basis is $C(\text{ann}(v))$.

Proof: Since $\deg \text{ann}(v) = d$, we can write it as

$$\text{ann}(v)(x) = x^d + \sum_{k=0}^{d-1} c_k x^k.$$

Thus

$$\begin{aligned} 0 &= \text{ann}(T)v = \left(T^d + \sum_{k=0}^{d-1} c_k T^k \right) v \\ &\Rightarrow T^d v = -1 \left(\sum_{k=0}^{d-1} c_k T^k v \right) \end{aligned}$$

Thus $\forall k \geq d$, $T^k v \in \text{span}(v, Tv, \dots, T^{d-1}v)$. In other words $Z(v, t)$ is generated by $\{v, \dots, T^{d-1}v\}$. Thus to prove that the set is a basis for V , it suffices to prove that they are linearly independent. Suppose

$$\sum_{k=0}^{d-1} a_k T^k v = 0.$$

Let

$$p(x) := \sum_{k=0}^{d-1} a_k x^k$$

$$p(T)v = 0 \Rightarrow p(T) \in \{f \in F[x] : f(T)v = 0\}.$$

$\text{ann}(v)$ generates the ideal so $\text{ann}(x) \mid p(x)$ which implies that $a_k = 0$ for all k (if not we have a polynomial $\deg p(x) < d$ in a ideal generated by a degree d polynomial). Thus the set is a basis for $Z(T, v)$. Let $w_i = T^i v$. Under the basis $\{w_0, \dots, w_{d-1}\}$ we know that $Tw_i = w_{i+1}$ if $i < d-1$ and if

$$\begin{aligned} Tw_{d-1} &= Tv_d \\ &= \sum_{k=0}^{d-1} (-c_k) w_k, \end{aligned}$$

as discussed above. Thus under this basis T is similar to the matrix $C(f)$. $\square$

Lemma 4.3. For $v, w \in V$ where $\gcd(\text{ann}(v), \text{ann}(w)) = 1$, $\text{ann}(v+w) = \text{ann}(v) \text{ann}(w)$

Proof: We know that $\text{ann}(v+w) \mid \text{ann}(v) \text{ann}(w)$ because polynomials in T commute.

$$\begin{aligned} \text{ann}(v+w)(T)(v+w) &= 0 \\ \text{ann}(v+w)(T)v &= -\text{ann}(v+w)(T)w \\ 0 &= \text{ann}(v)(T) \text{ann}(v+w)(T)v = -\text{ann}(v)(T) \text{ann}(v+w)(T)w \end{aligned}$$

Thus $\text{ann}(w) \mid \text{ann}(v) \text{ann}(v+w) \Rightarrow \text{ann}(w) \mid \text{ann}(v+w)$. You can get $\text{ann}(v) \mid \text{ann}(v+w)$ in a similar fashion. Thus $\text{ann}(v) \text{ann}(w) \mid \text{ann}(v+w)$. $\square$

Lemma 4.4 (Existence of a Vector of Maximal Order). There exists $v \in V$ with $\text{ann}(v) = m_T(x)$.

Proof: Note that $m_T(T) = 0$ (by definition), thus for all $v \in V$ we have $\text{ann}(v) \mid m_T$. Furthermore, note the following fact that $\text{ann}(v) \neq m_T$ if and only if $\ker \text{ann}(v)(T) \neq V$. Since $\mathbb{F}[x]$ is a PID we can write

$$m_T(x) = \prod_{i=1}^n (q_i(x))^{a_i}$$

where $\gcd(q_i, q_j) = 1$ for $i \neq j$. Now note the following $p_i = \frac{m_T}{q_i}$ is a polynomial of degree less than m_T and hence $\ker p_i(T) \neq V$. Thus there exists a w_i such that $p_i(T)w_i \neq 0$. Let

$$v_i = \frac{m_T(T)}{(q_i(T))^{a_i}} w_i$$

But then note that $(q_i(T))^{a_i-1}v_i = p_i(T)w_i \neq 0$ but $(q_i(T))^{a_i}v_i = 0$. Thus $\text{ann}(v_i) = (q_i)^{a_i}$. Now we have a set of vectors $v_1, \dots, v_n$ where $\text{ann}(v_i) = (q_i)^{a_i}$. Then $\text{ann}(v_1 + \dots + v_n) = m_T$ by Lemma 4.3. $\square$

Lemma 4.5 (Cyclic Complement Lemma). For v as in Lemma 4.4, there exists a T -invariant subspace $W \leq V$ with

$$V = Z(v, T) \oplus W.$$

Proof: Let $Z = Z(v, T)$. Consider $\mathcal{W}$, the set of T invariant subspaces $W \subset V$ such that $W \cap Z = \{0\}$. $\{0\} \in \mathcal{W}$, so its nonempty. V f.d so that means there exists $W \in \mathcal{W}$ such that $\dim W = \max(\dim U : U \in \mathcal{W})$. Suppose for contradiction, $V \neq Z \oplus W$. Then there exists $u \in V$ such that $u \notin Z \oplus W$. Consider the set

$$I = \{p \in F[x] : p(T)u \in Z \oplus W\}.$$

Note that $m_T \in I$. Thus I is a nonzero ideal of $\mathbb{F}[x]$, let f be its unique monic generator. By definition of $f \in I$, $f(T)u = z + w$ for $z \in Z$ and $w \in W$. Since $m_T \in I$, $f \mid m_T$. Thus $m_T = fg$ for $g \in \mathbb{F}[x]$.

$$0 = m_T(T)u = g(T)f(T)u = g(T)z + g(T)w.$$

Both Z and W are T -invariant hence $g(T)z \in Z$ and $g(T)w \in W$. Since intersection is trivial they must both be 0. Since $z \in Z(v, T)$, there is a polynomial $h \in \mathbb{F}[T]$ such that $z = h(T)v$. Thus $g(T)h(T)v = 0$. Because since v has maximal order ie $\text{ann}(v) = m_T$. That means $m_T \mid gh \Rightarrow gf \mid gh \Rightarrow f \mid h \Rightarrow h = f(x)q(x)$. Thus $z = f(T)q(T)v$. Let $u' = u - q(T)v$. Note that $u \notin Z \oplus W$ because $u \notin Z \oplus W$ and $q(T)v \in Z$. $f(T)u' = f(T)u - f(T)q(T)v = f(T)u - h(T)v = f(T)u - z = w \in W$. Let $W' = W + Z(u', T)$, which is a T invariant space and strictly contains W because $u' \notin W$ but it is in W' . Let $x \in W' \cap Z$, $x = w' + k(T)u'$. Thus

$$k(T)u' = x - w' \in Z \oplus W.$$

Since $u - u' = q(T)v \in Z$ (shown above), anything mapping u' to $Z \oplus W$ also maps u there as well. Thus $k \in I$ and $f \mid k \Rightarrow k = cf$. Thus

$$k(T)u' = f(T)c(T)u' = c(T)w \in W$$

Thus $x = w' + c(T)w \in W$. But $x \in Z \cap W \Rightarrow x = 0$. Thus $W' \cap Z = \{0\}$. This contradicts the assumption of the maximality of W . $\square$

Theorem 4.6 (Cyclic Decomposition Theorem). There exist $v_1, \dots, v_k \in V$ with

$$V = Z(v_1, T) \oplus \dots \oplus Z(v_k, T),$$

and, writing $a_i = \text{ann}(v_i)$, one has $a_{i+1} \mid a_i$ for $1 \leq i \leq k-1$.

Proof: It is true for a 1 dimensional vector space. Suppose its true for a vector space of dimension at least $k - 1$. Let V be a vector space of at dimension k . Let $v_1 \in V$ be such that $\text{ann}(v_1) = m_T$. Then $V = Z(v_1, T) \oplus W$ Lemma 4.5. Note the following $m_{T|_W} \mid m_T$. By the induction hypothesis $W = \bigoplus_i^m Z(w_i, T|_W)$. Basically done since $\text{ann}(w_i) \mid m_{T|_W} \mid \text{ann}(v_1)$. $\square$

Definition 4.7. The polynomials $a_1, \dots, a_k$ in Theorem 4.6 are the *invariant factors* of T .

Theorem 4.8 (Rational Canonical Form). There is a basis of V in which the matrix of T is block diagonal:

$$[T] = C(a_1) \oplus C(a_2) \oplus \dots \oplus C(a_k),$$

with $a_1, \dots, a_k$ the invariant factors of T .

Proof: By Theorem 4.6:

$$V = \bigoplus_{i=1}^k Z(v_i, T)$$

and each subspace is T -invariant. Thus $[T] = \bigoplus_{i=1}^k [T|_{Z(v_i, T)}] = \bigoplus_{i=1}^k C(\text{ann}(v_i))$. $\square$

Theorem 4.9 (Uniqueness of Invariant Factors). The invariant factors $a_1, \dots, a_k$ of T are uniquely determined by T (equivalently, by the similarity class of $[T]$). Moreover

$$a_1 = m_T(x), \quad a_1 a_2 \dots a_k = p_T(x).$$

Proof:

$\square$

Theorem 4.10 (Similarity Criterion). Let $A, B \in M_n(\mathbb{F})$. The following are equivalent:

- (a) A and B are similar over $\mathbb{F}$.
- (b) A and B have the same invariant factors.
- (c) $xI - A$ and $xI - B$ have the same Smith normal form over $\mathbb{F}[x]$.

Chapter 5

Matrix Exponential and Logarithm

Definition 5.1. For $A \in M_n(\mathbb{C})$, the *matrix exponential* is

$$e^A = \exp(A) = \sum_{k=0}^{\infty} \frac{A^k}{k!} \in M_n(\mathbb{C}).$$

The series converges absolutely (in any matrix norm) for every A .

5.1 Basic Properties of the Matrix Exponential

Theorem 5.2 (Elementary Properties). For all $A \in M_n(\mathbb{C})$:

- (a) $e^0 = I$.
- (b) e^A is invertible, with $(e^A)^{-1} = e^{-A}$.
- (c) If P is invertible, then $e^{PAP^{-1}} = Pe^AP^{-1}$.
- (d) $\det(e^A) = e^{\text{tr}(A)}$.

Proof: (A):

$$e^0 = \sum_{k=0}^{\infty} \frac{0^k}{k!} = 0^0 = I$$

(B): Suppose $(e^A)^{-1}$ is invertible.

$$\begin{aligned} e^{-A}e^A &= \left(\sum_{k=1}^{\infty} \frac{(-A)^k}{k!} \right) \left(\sum_{k=1}^{\infty} \frac{A^k}{k!} \right) \\ &= \sum_{n=1}^{\infty} \sum_{k=0}^n \left(\frac{(-A)^{n-k}}{(n-k)!} \right) \left(\frac{A^k}{k!} \right) \\ &= \sum_{n=1}^{\infty} \frac{1}{n!} \sum_{k=1}^n \frac{n!}{k!(n-k)!} A^k (-A)^{n-k} \\ &= \sum_{n=1}^{\infty} (A + (-A))^n = \exp(0) = I \end{aligned}$$

This is true for $\exp(A+B) = \exp(A)\exp(B)$

(C): Straightforward

(D): $A = PJP^{-1}$ where $J = D + N$ (diagonal + upper diagonal nilpotent) is a Jordan Normal Form Matrix.

$$\begin{aligned}\det(\exp(A)) &= \det(\exp(J)) \\ &= \det(\exp(D + N)) = \det(\exp(D) \exp(N))\end{aligned}$$

Let k be minimal where $N^k = 0$. Thus

$$\exp(N) = I + N + \frac{1}{2}N^2 + \dots + \frac{1}{k-1}N^{k-1}.$$

Note $\exp(N)$ is upper diagonal hence $\det(\exp(N)) = 1$. thus

$$\begin{aligned}\det(\exp(A)) &= \det(\exp(D)) = \det\left(\exp\left(\sum_{i=1}^n \lambda_i E_{ii}\right)\right) \\ &= \det\left(\prod_{i=1}^n \exp(\lambda_i E_{ii})\right) \\ &= \prod_{i=1}^n \det(\exp(\lambda_i E_{ii})) \\ &= \prod_{i=1}^n \exp(\lambda_i) = \exp(\operatorname{tr} D) = \exp(\operatorname{tr} A)\end{aligned}$$

□

Theorem 5.3 (Product Formula). If $AB = BA$, then $e^{A+B} = e^A e^B$. In particular, $e^A e^{-A} = I$.

Proof:

□

Remark 5.1. Commutativity is necessary: in general $e^{A+B} \neq e^A e^B$ when $[A, B] \neq 0$. The Baker–Campbell–Hausdorff formula describes $\log(e^A e^B)$ as an infinite series in nested commutators of A and B .

Theorem 5.4 (Derivative). For all $A \in M_n(\mathbb{C})$,

$$\frac{d}{dt} e^{tA} = A e^{tA} = e^{tA} A.$$

Proof:

□

5.2 Spectral Theory of the Matrix Exponential

Theorem 5.5 (Spectral Mapping). If $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A (counted with multiplicity), then $e^{\lambda_1}, \dots, e^{\lambda_n}$ are the eigenvalues of e^A (counted with multiplicity).

Proof:

□

Theorem 5.6 (Jordan Block Formula). Let $J_k(\lambda) = \lambda I + N$ where N is the $k \times k$ nilpotent shift. Then

$$e^{J_k(\lambda)} = e^\lambda \sum_{j=0}^{k-1} \frac{N^j}{j!} = e^\lambda \begin{pmatrix} 1 & 1 & \frac{1}{2!} & \cdots & \frac{1}{(k-1)!} \\ 0 & 1 & 1 & \cdots & \frac{1}{(k-2)!} \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Proof: Since λI and N commute, $e^{J_k(\lambda)} = e^{\lambda I} e^N = e^\lambda e^N$. The nilpotent series $e^N = \sum_{j=0}^{k-1} \frac{N^j}{j!}$ terminates because $N^k = 0$. □

Corollary 5.7. If $A = PJP^{-1}$ is the Jordan decomposition of A , then $e^A = Pe^JP^{-1}$, where $e^J = \text{diag}(e^{J_{k_1}(\lambda_1)}, \dots, e^{J_{k_r}(\lambda_r)})$.

5.3 Matrix Logarithm

Definition 5.8. A *logarithm* of $A \in M_n(\mathbb{C})$ is any $B \in M_n(\mathbb{C})$ satisfying $e^B = A$. The *principal logarithm* $\text{Log}(A)$ is the unique logarithm (when it exists) whose eigenvalues all lie in the strip $\{z \in \mathbb{C} : \text{Im}(z) \in (-\pi, \pi]\}$.

Theorem 5.9 (Existence of Complex Logarithm). Every invertible matrix $A \in M_n(\mathbb{C})$ has a logarithm. Equivalently, every $A \in \text{GL}_n(\mathbb{C})$ lies in the image of $\exp : M_n(\mathbb{C}) \rightarrow \text{GL}_n(\mathbb{C})$.

Proof:

□

Remark 5.2. Non-invertible matrices have no logarithm: $0 = \det(e^B) = e^{\text{tr}(B)} \neq 0$ for any B .

Theorem 5.10 (Principal Logarithm via Series). If $\|A - I\| < 1$ (for any submultiplicative matrix norm), then A is invertible and

$$\text{Log}(A) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} (A - I)^k.$$

Proof:

□

Theorem 5.11 (Non-Uniqueness of Logarithm). Unlike the scalar case, a matrix $A \in \text{GL}_n(\mathbb{C})$ may have uncountably many distinct logarithms. Specifically, if B is one logarithm of A and C commutes with A and satisfies $e^C = I$ (e.g. $C = 2\pi i k P$ for a suitable projection P), then $B + C$ is another logarithm of A .

Theorem 5.12 (Real Logarithm). A real invertible matrix $A \in \text{GL}_n(\mathbb{R})$ has a real logarithm (i.e. $B \in M_n(\mathbb{R})$ with $e^B = A$) if and only if every real negative eigenvalue of A has even algebraic multiplicity.

Proof:

□

Theorem 5.13 (Spectral Mapping for Logarithm). If $A \in \text{GL}_n(\mathbb{C})$ has eigenvalues $\mu_1, \dots, \mu_n$ (none zero), then the eigenvalues of $\text{Log}(A)$ are $\text{Log}(\mu_1), \dots, \text{Log}(\mu_n)$, where Log on the right denotes the principal branch of the scalar logarithm.

Corollary 5.14 (Inverse of Exponential). If all eigenvalues of $A \in M_n(\mathbb{C})$ lie in the strip $\{z : \text{Im}(z) \in (-\pi, \pi]\}$, then $\text{Log}(e^A) = A$.

Chapter 6

Selected Problems from Past UCLA Basic Exams

6.1 Diagonalization

Exercise 6.1 (F21 · 6). Let A be a complex $n \times n$ matrix such that $A^2 = A$.

1. Prove that A is similar to a diagonal matrix.
2. Prove that $\text{tr}(A)$ is a non-negative integer.

Exercise 6.2 (F21 · 7). Let A, B be complex $n \times n$ matrices with $AB = BA$. Prove that A and B have a common eigenvector.

Exercise 6.3 (F23 · 9). Show that two diagonalizable matrices are simultaneously diagonalizable if and only if they commute.

Exercise 6.4 (F24 · 2). Let $n \geq 2$ and let A be a normal complex $n \times n$ matrix with $\dim \ker(\lambda - A) \leq 1$ for all $\lambda \in \mathbb{C}$. Prove that any normal complex $n \times n$ matrix B that commutes with A takes the form

$$B = b_0 I + b_1 A + \cdots + b_{n-1} A^{n-1}$$

for some $b_0, \dots, b_{n-1} \in \mathbb{C}$.

Exercise 6.5 (F20 · 4). Let A be a 2×2 real matrix with eigenvalues 2 and -1 . Let X be the set of all 2×2 real matrices B such that the 4×4 block matrix

$$\begin{pmatrix} A & B \\ 0 & A \end{pmatrix}$$

is diagonalizable over $\mathbb{C}$. Prove that X is a 2-dimensional subspace of $M_2(\mathbb{R})$.

6.2 Jordan Canonical Form

Exercise 6.6 (F24 · 3). Let T be a linear operator on a finite-dimensional complex vector space with spectrum $\sigma(T) = \{0\}$. Show that T is nilpotent.

Exercise 6.7 (F24 · 4). Consider the polynomial

$$p(x) = x^4 + 2x^3 - 3x^2 - 4x + 4.$$

(Note: p vanishes only at $x = 1$ and $x = -2$.) Let A be a complex 4×4 matrix such that $p(A) = 0$. Assume also that $\text{Tr}(A) = 1$ and $\text{Rank}(A - I) = 2$. Determine the Jordan canonical form of A . Justify fully.

Exercise 6.8 (F20 · 3). Let M be a complex 4×4 matrix satisfying

$$M^6 = M^4 = 2M^3 - M^2.$$

Describe all possible Jordan canonical forms of M .

Exercise 6.9 (F22 · 3). Let A be an $n \times n$ real matrix such that $e^A = I$. Show, via consideration of the Jordan normal form, that the minimal polynomial of A has no repeated roots.

Exercise 6.10 (S22 · 3). Let A be the $n \times n$ shift matrix

$$A = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Find the Jordan normal form of A^2 . (Here n is an arbitrary positive integer.)

Exercise 6.11 (S22 · 4). Let A be a complex $n \times n$ matrix such that $A^2 = I_n$. Show that

$$\text{rank}(A + I_n) + \text{rank}(A - I_n) = n.$$

6.3 Square Roots and Spectral Theory

Exercise 6.12 (F21 · 10). Let V be a finite-dimensional inner product space over $\mathbb{C}$, and let T be a linear operator on V .

1. Prove that if T is invertible, then T has a square root, i.e., there exists a linear operator R on V such that $R^2 = T$.
2. Now assume T is diagonalizable but not necessarily invertible. Under what conditions does T have only finitely many distinct square roots, and how many are there?

Exercise 6.13 (F23 · 12). Let A be a non-singular complex $n \times n$ matrix. Show that there exists a non-singular complex $n \times n$ matrix B such that $B^2 = A$.

Chapter 6.3

References